

A certified computation of the maximum number of stable matchings of order five

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Abstract

Let $f(n)$ denote the maximum number of stable matchings of a stable-marriage instance with n men and n women. Determining $f(n)$ is a problem of Knuth (1976), restated as Open Problem #1 by Gusfield and Irving (1989). Eilers reported $f(5) = 16$ in 2022 (OEIS A357269) via a constraint-programming search; the computation produced no proof object and the accompanying paper has not appeared. We give an independent confirmation of $f(5) = 16$ by a different method (a compact SAT encoding), and, to our knowledge, the first machine-checkable certificate for the upper bound: an UNSAT proof of the statement “some 5×5 instance has at least 17 stable matchings,” emitted by `kissat` in DRAT format (1.09 GB) and verified by the independent checker `drat-trim`, which replayed 10,398,816 lemmas using 462,936,491 resolution steps and emitted an LRAT certificate. Total solver time was under ten minutes on a laptop. We outline the remaining steps toward a fully verified proof in Lean 4, and discuss the first open case, the conjecture $f(6) = 48$.

1 Introduction

A stable-marriage instance of order n consists of n men and n women, each equipped with a strict total preference order over the opposite side. A perfect matching μ is *stable* if there is no *blocking pair*: a man m and woman w , not matched to each other, who each prefer the other to their partner in μ . Gale and Shapley proved every instance has at least one stable matching. Let $f(n)$ be the maximum, over all instances of order n , of the number of stable matchings. Determining the behaviour of f is Research Problem #5 in Knuth’s 1976 monograph [1] and Open Problem #1 of Gusfield and Irving [2].

Asymptotically, the best bounds are $f(n) \geq 2.28^n$ (Thurber [3]) and $f(n) \leq 3.55^n$ (Palmer and Pálvölgyi [5], sharpening the first simply-exponential bound of Karlin, Oveis Gharan and Weber [4]). Exact values are recorded in OEIS A357269 [6]:

$$f(1), \dots, f(5) = 1, 2, 3, 10, 16.$$

The values through $f(4) = 10$ go back to Knuth’s example and an exhaustive search of Eilers, which also showed the extremal reduced instance of order 4 to be unique. The value $f(5) = 16$ was announced by Eilers in September 2022, obtained with a MiniZinc constraint-programming model; the OEIS entry cites a paper “in preparation” which, as far as we can determine, has

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not appeared, and the computation produced no independently checkable artifact—constraint solvers, unlike modern SAT solvers, emit no proof of exhaustiveness.

This note does two things.

1. **Independent confirmation.** We re-derive $f(5) = 16$ by a different route: a direct SAT encoding of “there exists an instance of order n with at least k stable matchings.” A solver finds a witness with 16 stable matchings in 0.1 seconds and refutes $k = 17$ in minutes. Two solvers with unrelated codebases (CaDiCaL via PySAT, and `kissat`) agree on the refutation.
2. **A certificate.** The `kissat` refutation was run with proof logging, producing a 1.09 GB DRAT proof, verified end-to-end by `drat-trim` [7] in 530 seconds. The resulting LRAT certificate (4.3 GB) is the format consumed by verified checkers, including those available in Lean 4 [8]. To our knowledge this is the first proof object of any kind for the upper bound $f(5) \leq 16$.

Comparable certified computations include the formal proof of the Ramsey number $R(4, 5) = 25$ [9] and the Lean-certified evaluation of $K_8(4, 2) = 23$ [10]. We regard the present note as the computational core of such a development for $f(5)$; the remaining formalization steps are outlined in Section 4.

2 The encoding

Fix n and k . We construct a CNF formula $F_{n,k}$ that is satisfiable if and only if some instance of order n has at least k stable matchings.

Preference variables. For each man m and each unordered pair $\{a, b\}$ of women, a variable p_{ab}^m meaning “ m prefers a to b ”; symmetrically for each woman. A strict total order is exactly a transitive tournament, and a tournament is transitive if and only if it contains no directed 3-cycle. So for every person and every unordered triple $\{a, b, c\}$ we add the two clauses forbidding the two cyclic orientations. For $n = 5$ this is 10 variables and 20 clauses per person.

Stability indicators. For each of the $n!$ perfect matchings μ , a variable S_μ with, for every non-matched pair (m, w) , the clause

$$S_\mu \rightarrow \neg(p_{w\mu(m)}^m \wedge p_{m\mu(w)}^w),$$

i.e., S_μ forces (m, w) not to block μ . Only this direction is needed:

Proposition 1. $F_{n,k}$ is satisfiable iff some instance of order n has $\geq k$ stable matchings.

Proof. ($\Leftarrow$) Given such an instance, set the preference variables accordingly and set S_μ true exactly for k of its stable matchings; every clause is satisfied. ($\Rightarrow$) A model’s preference variables define strict total orders (transitive tournaments), i.e., an instance; the cardinality constraint (below) forces at least k indicators true, and the implication clauses ensure each such μ has no blocking pair, i.e., is genuinely stable. $\square$

Cardinality. $\sum_\mu S_\mu \geq k$, encoded with a sequential counter.

Symmetry. For the refutation run we add unit clauses fixing man 1’s preference list to the identity order.

Lemma 1. *If some instance of order n has $\geq k$ stable matchings, then some instance in which man 1’s list is the identity order has $\geq k$ stable matchings.*

Proof. Relabel the women by the inverse of man 1’s preference permutation. Relabeling is a bijection on instances that maps stable matchings to stable matchings bijectively. $\square$

For $(n, k) = (5, 17)$ with the symmetry clauses, $F_{5,17}$ has 1,971 variables and 6,026 clauses.

3 Results

All runs on a single Apple M4 Max core (48 GB RAM, macOS 26.3).

Sanity check at $n = 4$. The encoder reproves $f(4) = 10$: $k = 10$ is satisfiable (witness recounted by an independent brute-force counter) and $k = 11$ is refuted, both in ≤ 0.1 seconds. The brute-force counter itself was validated against the extremal Latin instances of orders 3, 4, 5, 6 recorded in OEIS A351413 (values 3, 10, 9, 48, all reproduced).

Lower bound $f(5) \geq 16$. CaDiCaL 1.9.5 finds a model of $F_{5,16}$ in 0.1 seconds. Decoding yields the instance below (row i = preference list of man/woman i , most preferred first, participants numbered 1–5); the independent counter confirms it has exactly 16 stable matchings.

	2 3 1 5 4		4 1 3 2 5
	5 1 4 2 3		3 2 5 4 1
men:	4 5 1 3 2	women:	2 3 1 5 4
	3 2 4 5 1		5 1 2 3 4
	1 5 2 3 4		1 4 5 3 2

Upper bound $f(5) \leq 16$. $F_{5,17}$ (with Lemma 1 clauses) is unsatisfiable. Two independent refutations:

solver	time	proof
CaDiCaL 1.9.5 (PySAT)	389.6 s	none (plain run)
kissat 4.0.4	~ 8 min	binary DRAT, 1,093,088,497 bytes

drat-trim verified the DRAT proof by backward checking in 530.0 seconds: 10,398,816 proof lemmas, of which 6,686,848 in the trimmed core, checked with 462,936,491 resolution steps; 5,281 of the 6,026 input clauses are in the core; the core contains 2,490 RAT lemmas. The verdict was **s VERIFIED**, and an LRAT certificate of 4,327,129,882 bytes was emitted.

Theorem 1. $f(5) = 16$.

Proof. The instance displayed above has 16 stable matchings (checkable by enumerating the 120 perfect matchings). For the upper bound, by Proposition 1 and Lemma 1 it suffices that $F_{5,17}$ (with symmetry clauses) is unsatisfiable, which is witnessed by the verified DRAT/LRAT certificate. $\square$

4 Toward a Lean-verified proof, and $f(6)$

Three components remain for a formalization in the style of [9, 10]: (i) a Lean statement of Proposition 1 connecting $F_{5,17}$, as a concrete CNF object, to a Lean definition of stable matchings; (ii) Lemma 1; (iii) importing the LRAT certificate. For (iii) we note that the trimmed core contains 2,490 RAT lemmas, while Mathlib’s `lrat_proof` implements RUP-only checking; options include re-solving with RAT-introducing inprocessing disabled, converting the proof, or using a RAT-capable verified checker [8].

The first open case is now $n = 6$. The best known lower bound is $f(6) \geq 48$, attained by a Latin instance built from the dihedral group (OEIS A351413), and OEIS A357271 records the conjecture that 48 is exact. The search space of reduced instances grows from $\sim 10^{17}$ at $n = 5$ to $\sim 10^{28}$ at $n = 6$, so a direct extension of the present method will require substantially stronger symmetry breaking (only the $n!$ woman-relabeling symmetry is broken here; the men-relabeling and side-swap symmetries remain), or a structural route through rotation posets, whose downset counts equal the numbers of stable matchings.

Data availability. The encoder, counter, logs, DRAT and LRAT files, and instructions to reproduce every number in this note are available in the accompanying repository.

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